

- [Vectorized Portfolio Analytics Engine](#)

Vectorized Portfolio Analytics Engine

Financial analysts utilize advanced data processing libraries like `pandas` to monitor the health of complex investment portfolios. By performing vectorized operations on structured datasets, these systems can rapidly calculate gains, identify high-volatility assets, and generate relational reports that integrate disparate data sources like sector mappings.

Implement a `PortfolioAnalyzer` class that utilizes the `pandas` ecosystem for deep financial analytics. The engine must ingest holdings data from CSV files, calculate derived performance metrics (cost basis, market value, gain/loss percentage), and perform relational **Left Joins** to enrich the portfolio with categorical sector metadata.

Example 1 Input:

```
LOAD data/portfolio.csv
CALC
TOTAL
TOP
```

Output:

```
Total Value: 15700.0
Top: AAPL
```

Explanation: The analyzer loads the holdings (AAPL, TSLA, MSFT, AMZN, NFLX). It calculates the current market value of all shares. `AAPL` is identified as the top performer based on the highest percentage gain since purchase.

Example 2 Input:

```
UNDER -5
```

Output:

Under: NFLX

Explanation: The system filters for assets where the `gain_loss_pct` is less than or equal to -5%. In this case, `NFLX` is identified as an underperforming asset.

Function Parameter:

- `csv_path` (str): The filesystem path to the CSV data.
- `threshold_pct` (float): Benchmark for identifying underperforming assets.

Returns

- float: The aggregate market value.
- list: Symbols matching the performance criteria.

Constraints

- All calculations **must** be vectorized (no explicit Python loops over DataFrame rows).
- The `apply_sector_mapping` method must use a Left Join to ensure no portfolio data is lost during the merge.
- Formula for Gain/Loss %: $((market_value - cost_basis) / cost_basis) \times 100$.

Input Format for Custom Testing Input from stdin contains commands: `LOAD <file>`, `CALC`, `TOTAL`, `TOP`, `UNDER <threshold>`, or `SECTOR <file>`.

Sample Case 0 Sample Input

```
LOAD data/portfolio.csv
CALC
TOTAL
```

Sample Output

```
Total Value: 15700.0
```

Explanation The portfolio is loaded and the total market value is aggregated across all holdings.